APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree Examination

MODEL QUESTION PAPER — SET 1

Course Code: PCCST501 **Course Name:** Computer Networks

Maximum Marks: 60 **Duration:** 2 hours 30 minutes

PART A — Answer all questions. Each question carries 3 marks.

Q. No.	Question	Marks	CO
1	Explain the role of protocol layering in the design of the Internet.	3	CO1
2	Differentiate between client-server and peer-to-peer application paradigms.	3	CO1
3	Compare TCP and UDP with respect to connection establishment and service provided.	3	CO2
4	Explain the purpose of I/O multiplexing in TCP client/server applications.	3	CO2
5	Distinguish between unicast routing and multicast routing.	3	CO3
6	Explain how Quality of Service requirements influence network-layer packet delivery.	3	CO3
7	Describe the functions of the data-link layer in transferring data across a link.	3	CO4
8	Explain the difference between digital transmission and analog transmission.	3	CO5

PART B — Answer one full question from each module. Each question carries 9 marks.

Module 1 — Question 9 (CO: CO1; Marks: 9)

(a) Explain the functions of the application, transport, network, data-link and physical layers in Internet communication. (5 marks)
(b) Illustrate how HTTP and DNS support a user accessing a website using its domain name. (4 marks)

Module 1 — Question 10 (CO: CO1; Marks: 9)

(a) Compare the client-server and peer-to-peer paradigms with respect to resource distribution, scalability and communication. (5 marks)
(b) Explain the operation of BitTorrent as a peer-to-peer file-sharing system. (4 marks)

Module 2 — Question 11 (CO: CO2; Marks: 9)

(a) Compare TCP and UDP in terms of connection establishment, reliability, flow control and overhead. (5 marks)
(b) A TCP sender transmits three consecutive segments with sequence numbers 1000, 1200 and 1400, each carrying 200 bytes. Determine the next expected byte sequence number after all three segments are received in order. (4 marks)

Module 2 — Question 12 (CO: CO3; Marks: 9)

(a) Explain the functions of unicast routing and the distinction between intra-domain and inter-domain routing. (5 marks)
(b) A router has the following link costs: A–B = 2, A–C = 5, B–C = 1, B–D = 4, C–D = 1. Compute the minimum-cost path from A to D and its total cost. (4 marks)

Module 3 — Question 13 (CO: CO4; Marks: 9)

(a) Explain the role of MAC protocols in coordinating access to a shared communication medium. (5 marks)
(b) Distinguish between the functions of a switch and a router when connecting network devices. (4 marks)

Module 3 — Question 14 (CO: CO4; Marks: 9)

(a) Explain the operation of the Ethernet protocol and the role of link-layer addressing in frame delivery. (5 marks)
(b) Illustrate how a wireless LAN differs from a wired Ethernet LAN in medium access and transmission. (4 marks)

Module 4 — Question 15 (CO: CO5; Marks: 9)

(a) Explain the relationship between data, signals and transmission media in the physical layer. (5 marks)
(b) A digital link transmits 2400 bits in 0.8 seconds. Compute its data rate in bits per second. (4 marks)

Module 4 — Question 16 (CO: CO5; Marks: 9)

(a) Compare analog transmission and digital transmission with respect to signal representation and susceptibility to noise. (5 marks)
(b) Explain how bandwidth utilization and transmission-medium characteristics affect communication performance. (4 marks)

— End of Question Paper —

B. BLUEPRINT TABLE

Q. No.	Module	Syllabus topic	CO	Bloom's level	Marks
1	1	Protocol layering	CO1	Understand	3
2	1	Application-layer paradigms	CO1	Understand	3
3	2	Transport-layer services and protocols: TCP, UDP	CO2	Understand	3
4	2	TCP client/server; I/O multiplexing	CO2	Understand	3
5	3	Unicast and multicast routing	CO3	Understand	3
6	3	Quality of Service	CO3	Understand	3
7	3	Data-link layer functions	CO4	Understand	3
8	4	Digital and analog transmission	CO5	Understand	3
9	1	Protocol layering; HTTP and DNS	CO1	Apply	9
10	1	Client-server, P2P networks, BitTorrent	CO1	Analyse	9
11	2	TCP/UDP; TCP sequence-number trace	CO2	Apply	9
12	2	Unicast routing; intra-domain and inter-domain routing	CO3	Apply	9
13	3	MAC protocols; connecting devices	CO4	Apply	9
14	3	Ethernet, link-layer addressing, wireless LANs	CO4	Analyse	9
15	4	Data and signals; transmission media; data rate	CO5	Apply	9
16	4	Analog/digital transmission; bandwidth utilization	CO5	Analyse	9

C. VERIFICATION CHECKLIST

Check	Result
Part A marks	8 questions × 3 marks = 24 marks
Part B marks	4 questions answered × 9 marks = 36 marks
Module distribution	Part A: Q1–Q2 Module 1; Q3–Q4 Module 2; Q5–Q6 Module 3; Q7–Q8 Module 4
Part B alternatives	Module 1: Q9/Q10; Module 2: Q11/Q12; Module 3: Q13/Q14; Module 4: Q15/Q16
Subdivision totals	Q9–Q16 each total 9 marks; no question has more than two subdivisions
CO coverage	CO1, CO2, CO3, CO4 and CO5 are assessed
Problem-solving	Q11, Q12 and Q15 are numerical/protocol-trace problems; Q9 and Q13 also require application
Numerical checks	Q11: 1000 + 3×200 = 1600. Q12: A–B–C–D costs 2+1+1 = 4, the minimum. Q15: 2400/0.8 = 3000 bits/s.
No earlier-set repetition	No previous sets have been issued in this sequence; this is the baseline Set 1.

Official Course Outcomes supplied

CO1: Understand the internetworking design in terms of protocol stack and the role of various application layer protocols

CO2: Illustrate the functions of the transport layer from connectionless and connection-oriented perspectives

CO3: Identify how the network layer achieves host-to-host connectivity and caters to the diverse service requirements of the host applications

CO4: Explain the nuances of the data link layer design and demonstrate the various data link link layer protocols

CO5: Describe the fundamental characteristics of the physical layer and understand how the physical layer supports the functionalities of the top layers

D. SCHEME OF VALUATION

Part A

Q1: Protocol layering divides communication into layers (1); each layer has defined functions (1); coordinated protocols/interfaces support communication (1).

Q2: Client-server has service-providing servers (1); P2P distributes roles among peers (1); valid comparison of resource distribution/scalability (1).

Q3: TCP connection-oriented and UDP connectionless (1); TCP reliable/ordered delivery (1); UDP lower overhead without TCP reliability guarantees (1).

Q4: Defines I/O multiplexing (1); monitors multiple descriptors/connections (1); enables efficient handling of multiple clients/events (1).

Q5: Unicast is one-to-one (1); multicast is one-to-many group delivery (1); valid routing/destination distinction (1).

Q6: Names QoS needs such as delay, bandwidth or loss (1); identifies relevant policy/mechanism (1); links it to packet delivery (1).

Q7: Framing (1); link-layer addressing (1); medium access or error detection (1).

Q8: Digital uses discrete levels (1); analog varies continuously (1); valid noise/regeneration comparison (1).

Part B

Q9(a) — 5: Application services (1); transport process-to-process delivery (1); network host-to-host routing (1); data-link frame delivery (1); physical bit transmission (1).

Q9(b) — 4: HTTP requests/receives web resources (2); DNS resolves domain name to IP address (2).

Q10(a) — 5: Resource distribution (2); scalability (2); communication/coordination difference (1).

Q10(b) — 4: File divided into pieces (1); peers exchange pieces (1); multiple peers contribute (1); reduces reliance on a single source (1).

Q11(a) — 5: Connection establishment (1); reliability/order (2); flow control (1); overhead (1).

Q11(b) — 4: Starting sequence number 1000 (1); total payload = 3×200 = 600 bytes (2); next expected byte = 1600 (1).

Q12(a) — 5: Unicast routing (1); forwarding table/next hop (2); intra-domain routing (1); inter-domain routing (1).

Q12(b) — 4: Valid path A–B–C–D (1); cost calculation 2+1+1 (2); minimum cost 4 (1).

Q13(a) — 5: Shared-medium access issue (1); MAC coordination purpose (1); access method (1); collision handling/avoidance where applicable (1); efficient sharing (1).

Q13(b) — 4: Switch forwards frames using link-layer information (2); router forwards packets using network-layer information (2).

Q14(a) — 5: Ethernet frame transmission (2); source/destination MAC addresses (1); LAN frame delivery (1); relevant Ethernet protocol characteristic (1).

Q14(b) — 4: Wired medium (1); wireless radio transmission (1); access coordination comparison (1); consequence for interference/transmission (1).

Q15(a) — 5: Data represents information (1); signals encode data (2); medium carries signals (2).

Q15(b) — 4: Rate formula bits/time (1); substitution 2400/0.8 (2); answer 3000 bits/s (1).

Q16(a) — 5: Analog representation (1); digital representation (1); analog noise effect (1); digital noise/regeneration (1); further valid comparison (1).

Q16(b) — 4: Bandwidth/capacity relationship (2); medium characteristics (1); performance effect (1).